

Customer Segmentation and Segment Specific Product Association Mining

Online Retail Customer Analytics



Sri Lanka Institute of Information Technology

IT3091 Machine Learning

Guided Data Track

Group 2026 DS 10

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Executive Summary

This project identifies commercially meaningful customer groups and product relationships for a UK-based online retailer. The decision need is practical: determine which customers should receive loyalty, retention or reactivation treatment, and which products should be promoted together within each group. The analysis uses the UCI Online Retail transaction dataset and connects customer-level clustering to invoice-level association mining.

After documented quality controls, 391,150 customer-linked merchandise rows representing 4,334 customers and 18,402 invoices were used for segmentation and segment-specific basket analysis. A five-cluster K-Means model using log-transformed Recency, Frequency and Monetary features with StandardScaler was selected. It achieved a silhouette score of 0.319, Davies-Bouldin index of 0.983 and strong stability (mean seed ARI 0.985; mean resampling ARI 0.931). The solution balanced statistical quality with usable cluster sizes and business interpretation.

The five segments are Hibernating Low Value, Recent Low Engagement, At Risk Regular, Loyal High Value and Champions. Champions represent only 8.26% of customers but 53.83% of revenue, while Hibernating Low Value customers represent 26.21% of customers and 3.50% of revenue. Segment-specific FP-Growth produced 2,361 practical rules after multi-metric filtering. Forty diverse recommendations were retained, eight per segment, for targeted bundles, cross-selling and lifecycle campaigns.

The principal recommendation is to deploy segment-specific rather than uniform marketing: protect Champions through VIP retention and premium bundles; grow Loyal High Value customers through relevant cross-sells; reactivate At Risk Regular customers using familiar product families; use low-cost onboarding prompts for Recent Low Engagement customers; and apply selective win-back offers to Hibernating Low Value customers. These are decision-support insights, not causal guarantees, and should be A/B tested before broad deployment.

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1 Business Problem and Analytical Design

1.1 Stakeholder and decision need

The stakeholder is a marketing and merchandising team at a UK-based non-store online retailer. Historical transactions contain evidence about purchase recency, purchasing occasions, customer value and products bought together, but they do not directly state which customers require retention or which bundles should be promoted. The project therefore converts transaction history into an interpretable decision structure.

1.2 Lenses tasks and outputs

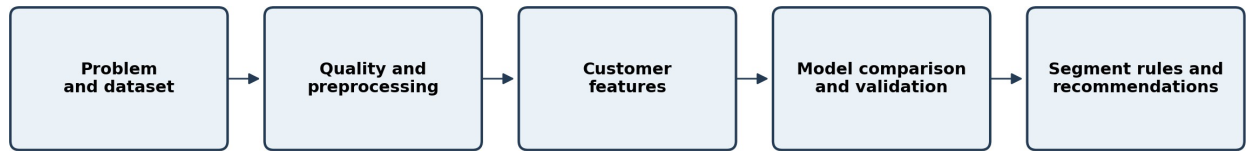
Element	Definition
Primary lens	Customer segmentation for lifecycle marketing and value management
Secondary lens	Product association and bundling within each final segment
Segmentation unit	One identified customer
Association unit	One valid purchase invoice represented by product presence
Primary output	Validated five-segment customer assignment and segment profiles
Secondary output	Ranked segment-specific product recommendations
Decision question	Which segments merit loyalty, retention or reactivation action, and which products should be cross-sold within each segment?

1.3 Objectives and success criteria

- Preserve raw evidence while creating task-specific valid purchase datasets.
- Create reproducible customer features with explicit units and leakage controls.
- Compare an interpretable RFM baseline and at least three clustering alternatives.
- Select a model using internal metrics, stability, cluster balance and business meaning rather than a single score.
- Mine robust, interpretable product rules separately for each segment.
- Translate results into actionable recommendations with risks and validation needs.

2 Workflow and Decision Governance

Figure 1 End to end analytical workflow



Decision log, fingerprints, reload checks and peer review across every stage

The workflow deliberately separates customer modelling from basket modelling. Customer records are aggregated only after purchase validation; association mining returns to invoice-product presence and then attaches the validated segment assignment. This prevents a row-level transaction from being mistaken for either a customer or a basket.

2.1 Team responsibilities

Member	ID	Core responsibility
Lakshitha Dilshan J.K.P.	IT24102621	Data understanding, preprocessing and quality assurance
Nirmani K.H.D.T.	IT24103380	Feature engineering, transformation and handover
Palliyaguruge C.T.	IT24103655	Clustering, model comparison, stability and profiling
Senaratne P.A.R.T.	IT24103526	Association mining, rule evaluation and recommendations

2.2 Decision governance

Thirty-eight decisions were recorded with the alternatives considered, selection rationale, evidence, impact, responsible member and reviewer. Examples include preserving a task-specific basket dataset, retaining genuine extreme customers while using transformations, rejecting high-silhouette but operationally unusable models, selecting five clusters for actionable detail, and validating association rules across four support thresholds. This log makes judgement visible rather than presenting the final method as inevitable.

3 Data Understanding and Quality

3.1 Dataset identity

Attribute	Value
Dataset	Online Retail
Provider	UCI Machine Learning Repository
Period	1 December 2010 to 9 December 2011
Raw rows	541,909 transaction lines
Fields	InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country
Geography	Transactions from multiple countries; UK dominates
Raw-file handling	Original file preserved; identity verified with SHA-256 fingerprint

3.2 Row meaning and variable roles

Variable	Role	Data quality relevance
InvoiceNo	Invoice identifier	C prefix signals cancellation; repeated values are valid line items
StockCode	Product identifier	Preferred to variable descriptions; reviewed administrative codes separately
Description	Product label	Missing and inconsistent text limits interpretation
Quantity	Count	Negative values indicate returns or adjustments
InvoiceDate	Timestamp	Final month is partial; supports recency and time analysis
UnitPrice	Numeric value	Non-positive prices cannot represent positive purchase revenue
CustomerID	Customer identifier	Missing IDs cannot be reliably assigned or imputed
Country	Category	Strong UK imbalance limits geographic generalisation

3.3 Material data issues and implications

Issue	Evidence	Analytical response
Missing customer identity	135,080 rows (24.93%)	Exclude from customer segmentation and segment-linked rules; do not fabricate IDs
Exact duplicates	5,268 additional rows (0.97%)	Retain one exact copy; do not remove repeated invoice numbers generally
Cancellations and returns	9,288 cancellation rows; 10,624 negative-quantity rows	Preserve separately; exclude from positive purchase models
Non-positive prices	2,517 rows	Exclude from positive revenue calculations
Product naming variation	217 StockCodes have multiple cleaned descriptions	Use StockCode as stable product key and canonical descriptions for reporting
Extreme valid purchases	Top 1% of customers generate 32.20% of revenue	Retain, transform, scale and test stability rather than delete automatically

Issue	Evidence	Analytical response
Partial final month	Data ends 9 December 2011	Retain for recency but avoid treating it as a complete month
Geographic imbalance	UK: 91.45% of valid rows; 84.59% of valid revenue	Do not make broad geographic claims; consider UK-only sensitivity later

These controls reflect the task: missing CustomerID is acceptable for overall merchandise basket analysis but not for customer segmentation. Similarly, administrative codes may affect billed value yet are unsuitable as merchandise products. Separate datasets preserve useful evidence without forcing one universal cleaning rule onto incompatible analytical units.

4 Preprocessing and Feature Engineering

4.1 Purchase validation audit

Step	Rule	Removed	Rows after
C01	Remove additional exact duplicate rows	5,268	536,641
C02	Exclude C-prefixed cancellation invoices	9,251	527,390
C03	Retain rows with Quantity greater than zero	1,336	526,054
C04	Retain rows with UnitPrice greater than zero	1,176	524,878
C05	Retain valid purchases with an identified CustomerID	132,186	392,692
C06	Exclude reviewed administrative StockCodes from the primary customer merchandise dataset	1,542	391,150

The final primary customer merchandise dataset contains 391,150 positive, customer-linked merchandise lines. A broader merchandise basket dataset retains transactions without CustomerID where invoice-product association remains meaningful. Returns, adjustments and reviewed administrative records are retained separately for audit and sensitivity analysis.

4.2 Feature design

Feature	Definition	Use
Recency	Days between 10 December 2011 and latest valid purchase	Core clustering feature
Frequency	Distinct valid purchase invoices per customer	Core clustering feature
Monetary	Sum of positive merchandise line value	Core clustering feature
ProductDiversity	Distinct purchased StockCodes	Enhanced candidate and profiling
AverageBasketSize	Mean merchandise units per purchase invoice	Enhanced candidate and profiling
TotalQuantity	Total merchandise units	Profiling; excluded from core due redundancy
AverageOrderValue	Mean value per invoice	Profiling
TenureDays	First-to-latest purchase span	Profiling; strongly related to frequency
AverageDaysBetweenPurchases	Mean gap for repeat customers	Profiling; missing by definition for 1,505 one-purchase customers

4.3 Transformation scaling leakage and outliers

RFM and behavioural measures were strongly right-skewed: Monetary skewness was 19.58 and ProductDiversity skewness was 6.92. Log1p transformations reduced those values to approximately 0.40 and -0.24 respectively, reducing domination by genuine wholesale customers without deleting them. Raw and transformed feature sets were retained so the effect could be compared rather than assumed.

Scaling was performed inside each clustering configuration. StandardScaler and RobustScaler were compared rather than storing one permanently scaled dataset. Customer aggregation used only records on or before the dataset end, and no downstream cluster label or recommendation was used to construct input features. Because this is unsupervised historical segmentation rather than predictive forecasting, conventional label leakage does not apply; nevertheless, task boundaries and the fixed analysis date prevent future information or output labels from entering the inputs.

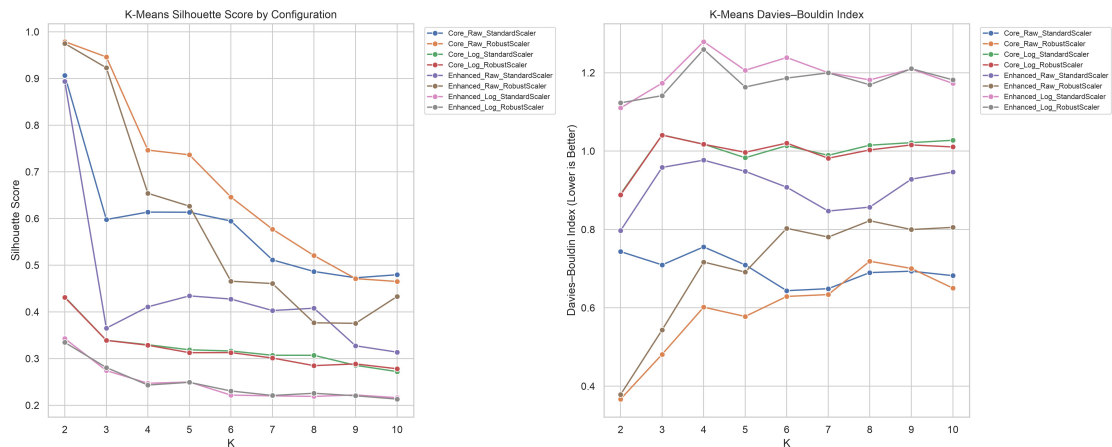
5 Modelling Strategy and Comparison

5.1 Baseline and alternatives

Method	Role	Key assumption or trade-off
Quartile RFM scoring	Interpretable baseline	Transparent but imposes discretised thresholds and limited interaction structure
K-Means	Centroid-based candidate	Efficient and interpretable; sensitive to scale, skew and k
Agglomerative clustering	Hierarchical alternative	Can reveal nested structure; linkage choice matters
Gaussian mixture model	Probabilistic alternative	Allows soft membership; assumes mixture distributions
DBSCAN	Density-based alternative	Finds noise and irregular groups; sensitive to epsilon and density variation

The experiment grid covered core and enhanced feature sets, raw and log-transformed representations, two scalars, K-Means k values, alternative algorithms and DBSCAN parameter combinations. In total, 192 unique clustering experiments were evaluated. This exceeds a nominal three-alternative comparison and exposes sensitivity to both algorithm choice and preprocessing.

Figure 2 K-Means configuration comparison



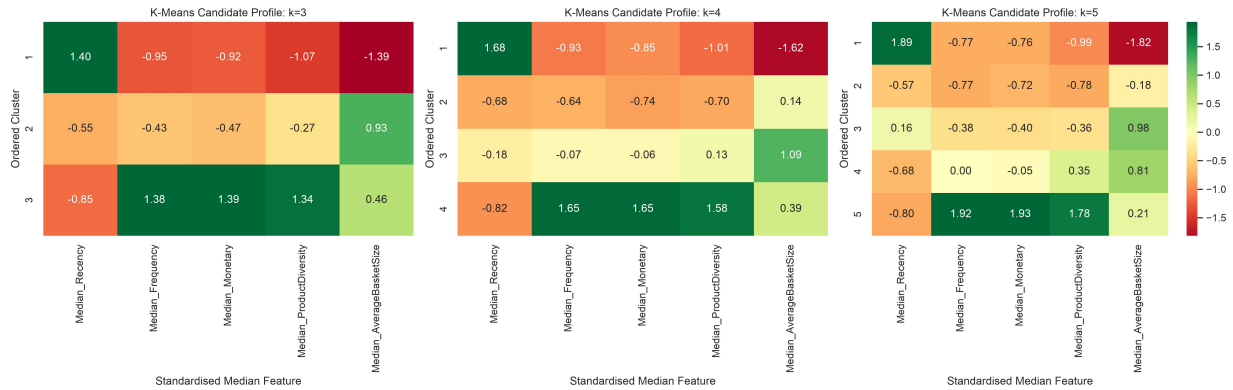
5.2 Critical model selection

The highest silhouette score was not automatically selected. Raw-feature solutions obtained apparently excellent separation by isolating tiny groups of extreme customers; one placed 99.82% of customers in a single cluster. Such a solution is statistically attractive but commercially unusable. The team therefore combined silhouette, Davies-Bouldin, Calinski-Harabasz, cluster-size balance, noise percentage, stability and interpretability.

Candidate	k	Silhouette	DB index	Smallest	Largest	Seed ARI	Resample ARI
KMeans_Core_Log_Standard_k3	3	0.339	1.041	17.4%	43.1%	0.996	0.974
KMeans_Core_Log_Standard_k4	4	0.330	1.018	16.1%	36.9%	0.981	0.943
KMeans_Core_Log_Standard_k5	5	0.319	0.983	8.3%	26.2%	0.985	0.931

The k=3 candidate had the highest silhouette and strongest stability, but k=5 retained competitive validation metrics while providing finer, actionable lifecycle distinctions and cluster sizes between 8.26% and 26.21%. The selected configuration was therefore K-Means with five clusters on log-transformed core RFM features using StandardScaler, random_state 42 and n_init 20.

Figure 3 Candidate profile comparison supporting the five-cluster choice



5.3 Selected model metrics

Metric	Result	Interpretation
Silhouette	0.319	Moderate separation in behavioural data
Davies-Bouldin	0.983	Approximately one; assessed with other criteria
Calinski-Harabasz	3,231.0	Useful relative compactness and separation indicator
Mean seed ARI	0.985	Assignments highly consistent across random seeds
Mean resampling ARI	0.931	Strong consistency over repeated 80% samples
Minimum resampling ARI	0.818	Worst tested sample remained substantially aligned
RFM baseline ARI	0.361	Only moderate agreement with quartile labels
RFM baseline NMI	0.488	Related but not equivalent information structure

6 Evaluation Validation and Critical Judgement

6.1 Stability and sensitivity

Ten random seeds tested initialization sensitivity and thirty repeated 80% customer samples tested whether moderate sample changes altered the segmentation. Adjusted Rand Index was used because it measures agreement between assignments while correcting for chance. The selected model achieved a mean seed ARI of 0.985 and mean resampling ARI of 0.931, with a minimum resampling ARI of 0.818.

Figure 4 Candidate stability across random seeds

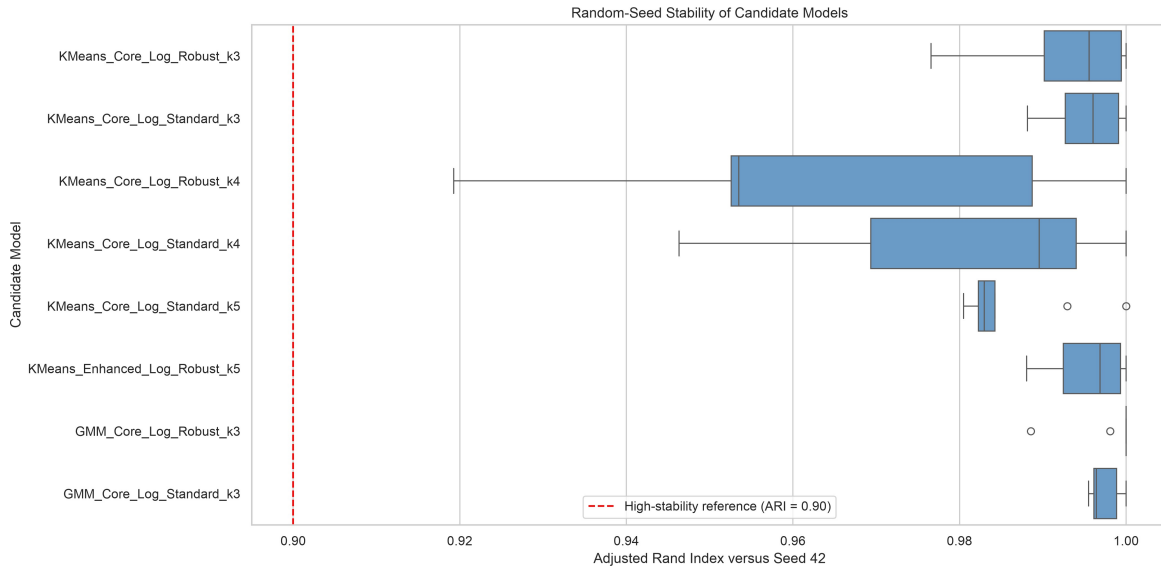
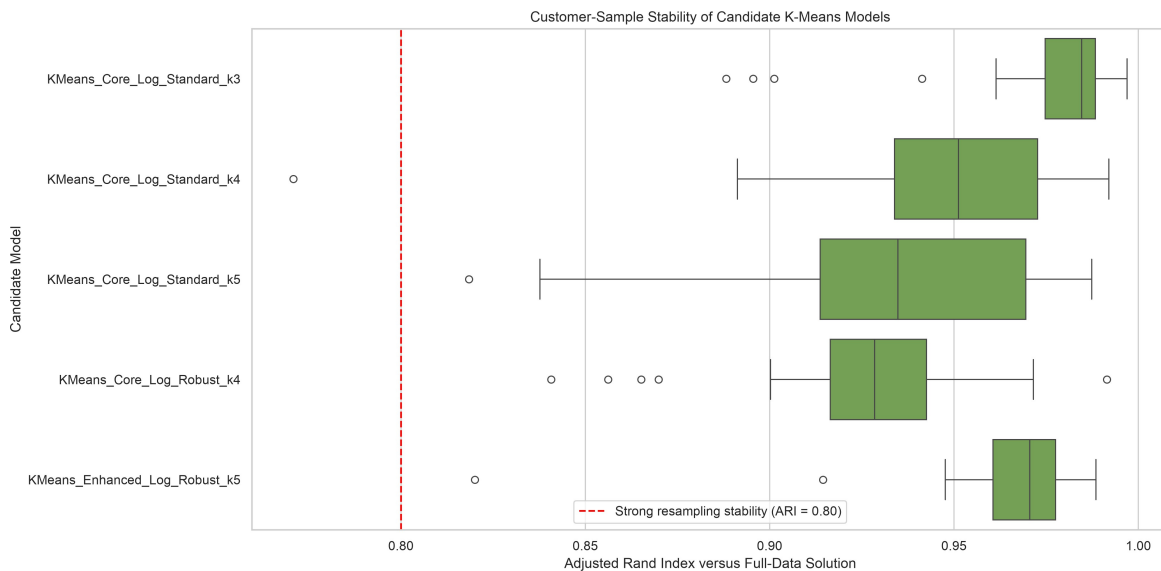


Figure 5 Candidate stability under customer resampling



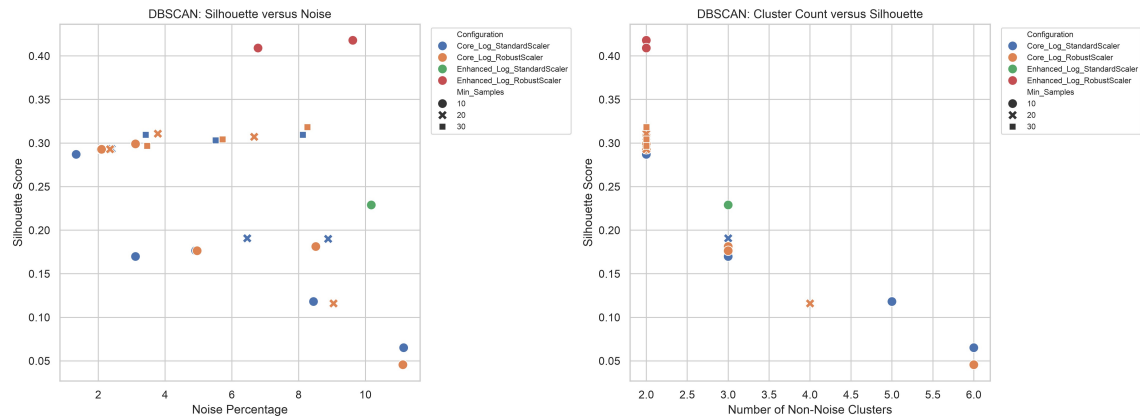
6.2 Honest interpretation

A silhouette score of 0.319 does not indicate perfectly separated natural classes. Retail behaviour is continuous, and customers near cluster boundaries may reasonably resemble more than one segment. The value of the five-cluster solution lies in its reproducibility, operational balance and interpretable profiles, not a claim that five objectively true customer types exist. Cluster labels are descriptive names assigned after original-scale profiling; they are not observed ground-truth outcomes.

No train-test accuracy is reported because clustering and association mining are unsupervised descriptive tasks. Validation instead uses stability, preprocessing sensitivity, alternative algorithms, cluster balance, association-

rule support experiments and transparent thresholding. Future business evaluation should add out-of-time validation and controlled campaign experiments.

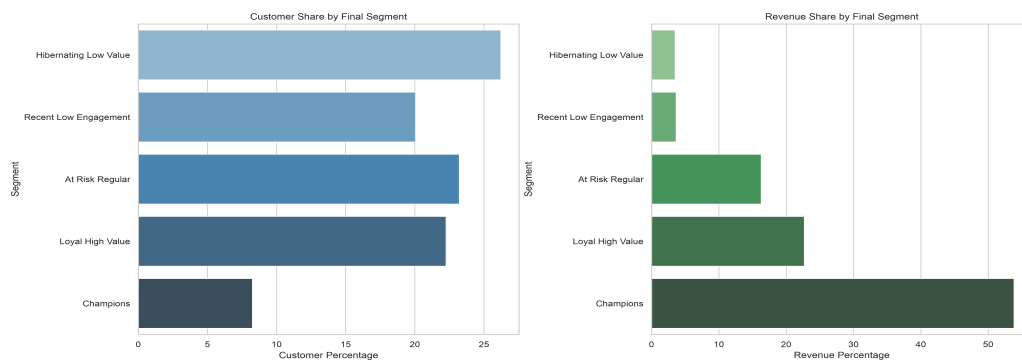
Figure 6 DBSCAN parameter search and density-based comparison



7 Segment Profiles and Business Value

Segment	Customers	Customer share	Revenue share	Median R	Median F	Median M
Hibernating Low Value	1,136	26.21%	3.50%	219	1	GBP 239
Recent Low Engagement	869	20.05%	3.67%	26	1	GBP 336
At Risk Regular	1,006	23.21%	16.28%	83	3	GBP 981
Loyal High Value	965	22.27%	22.71%	17	5	GBP 1,717
Champions	358	8.26%	53.83%	8	15	GBP 5,795

Figure 7 Customer share compared with revenue share



7.1 Segment interpretations

Hibernating Low Value. Large inactive group with low frequency and value. Broad expensive incentives are unlikely to be efficient; use selective win-back tests and low-cost reminders.

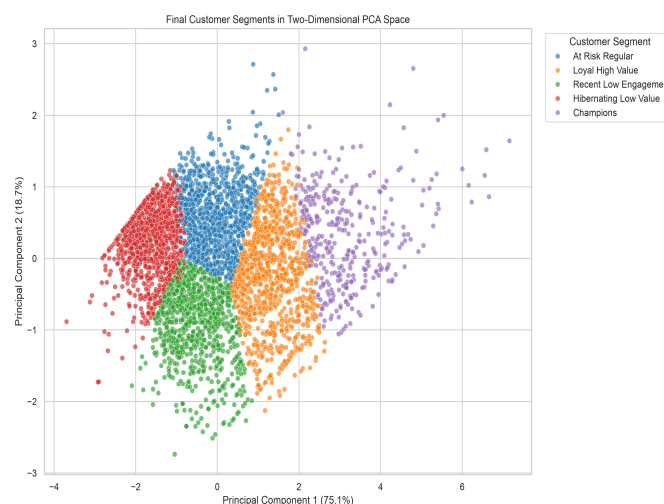
Recent Low Engagement. Recently active but mostly single-purchase customers. Use onboarding, second-purchase prompts and simple complementary offers while interest is current.

At Risk Regular. Previously regular and meaningful-value customers with deteriorating recency. Prioritise personalised reactivation using familiar categories and time-limited bundles.

Loyal High Value. Recent repeat purchasers with substantial value and tenure. Reinforce loyalty, cross-sell relevant ranges and prevent migration into the at-risk group.

Champions. Smallest group but dominant revenue contributor. Protect through VIP service, premium access and carefully targeted high-confidence bundles; avoid unnecessary discounting.

Figure 8 Two-dimensional PCA view of final customer segments



8 Segment Specific Association Mining

8.1 Basket construction and FP-Growth

Each valid segment-linked invoice was encoded as a sparse Boolean product-presence basket. Duplicate invoice-product pairs were collapsed, so multiple units of the same product did not inflate co-occurrence. The five baskets contained 1,364 to 7,183 invoices and densities below 0.72%, which supported sparse FP-Growth. A maximum itemset length of three maintained interpretability.

The initial minimum support was adaptive: $\max(1\%, 20 / \text{segment invoices})$. Rules initially required confidence of at least 0.30 and lift of at least 1.20. Practical filtering then required at least 25 supporting invoices, confidence at least 0.40, lift at least 1.50 and positive leverage. This combination counters the tendency of confidence to favour popular consequents and lift to exaggerate rare relationships.

Segment	Invoices	Practical rules	Median count	Median confidence	Median lift	Max lift
Hibernating Low Value	1,365	19	30	0.694	12.77	25.79
Recent Low Engagement	1,364	26	32	0.570	12.92	26.33
At Risk Regular	3,073	550	37	0.586	18.83	72.67
Loyal High Value	5,417	839	69	0.559	10.72	70.04
Champions	7,183	927	91	0.551	9.41	53.81

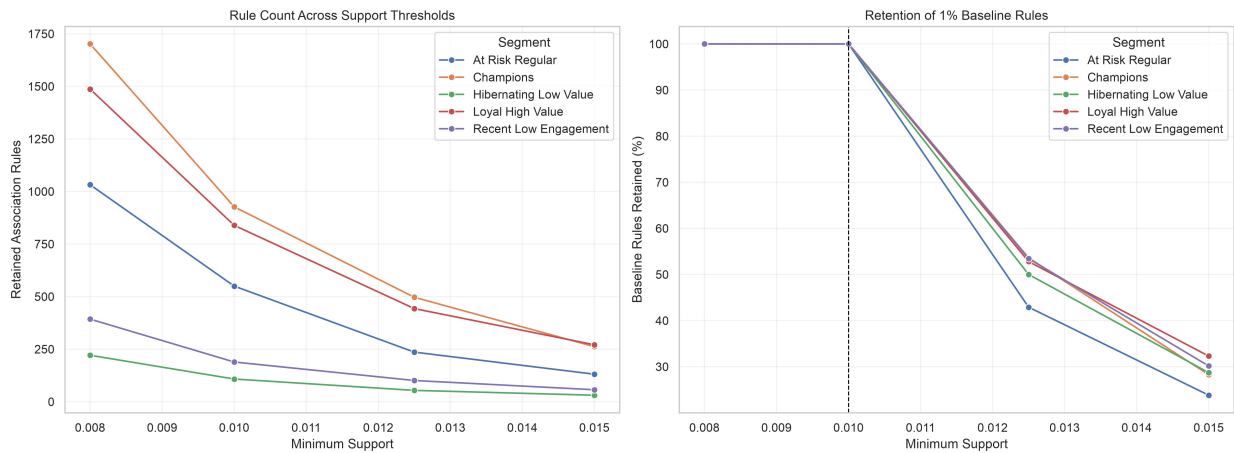
Figure 9 Association-rule comparison across customer segments



8.2 Support sensitivity and recommendation selection

Support thresholds of 0.008, 0.010, 0.0125 and 0.015 were compared for every segment, producing 20 experiments. Lower support substantially increased rule volume and maximum lift but also increased the risk of unstable rare patterns. At stricter thresholds, approximately 24% to 54% of the 1% baseline rules remained, depending on segment. The 1% baseline was retained as a balance between coverage and evidence, followed by the absolute 25-invoice practical filter.

Figure 10 Association-rule support sensitivity



The initial process retained 3,129 rules; 2,361 passed practical filters. A percentile-based rule score balanced support, confidence, lift and leverage. To avoid a final list dominated by multiple rules recommending the same item, eight high-quality rules with distinct consequent products were selected for each segment, resulting in 40 recommendations.

8.3 Illustrative recommendations

Segment	If basket contains	Recommend	Confidence	Lift	Channel
At Risk Regular	HERB MARKER BASIL	HERB MARKER MINT	0.977	66.74	Personalised email or direct campaign
Champions	GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.826	16.95	VIP communication and premium campaigns
Hibernating Low Value	GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.735	18.92	Email or low-cost digital campaign
Loyal High Value	WOODEN STAR CHRISTMAS SCANDINAVIAN	WOODEN HEART CHRISTMAS SCANDINAVIAN	0.794	24.99	Loyalty programme and personalised offers
Recent Low Engagement	WOODEN TREE CHRISTMAS SCANDINAVIAN	WOODEN STAR CHRISTMAS SCANDINAVIAN	0.843	12.92	Post-purchase email or website recommendation

9 Recommendations Limitations and Responsible Use

9.1 Stakeholder action plan

Segment	Priority action	Measurement
Champions	VIP retention, premium access and high-confidence complementary bundles	Repeat rate, margin, retention and bundle attachment
Loyal High Value	Relevant cross-selling and loyalty recognition	Frequency, basket value and segment migration
At Risk Regular	Personalised reactivation using previously purchased product families	Reactivation and incremental revenue versus control
Recent Low Engagement	Second-order prompts and low-friction bundles	Second-purchase conversion within 30 to 60 days
Hibernating Low Value	Selective low-cost win-back campaigns; suppress after repeated non-response	Cost per reactivated customer and unsubscribe rate

Deployment should start as controlled pilots. For each segment, randomly assign eligible customers to treatment and control groups, hold the targeting window fixed, and measure incremental conversion, revenue, margin and opt-out rates. Product rules should be refreshed periodically and blocked where inventory, margin, seasonality or brand constraints make the bundle unsuitable.

9.2 Limitations and risks

- 9.2.1 The dataset covers roughly one year for one retailer and ends with an incomplete month; results may not generalise across periods, retailers or economic conditions.
- 9.2.2 Missing CustomerID excludes 24.93% of raw rows from customer-level modelling, so anonymous buyers are not represented in segments.
- 9.2.3 The UK dominates the data. Segment meaning and product relationships may differ in smaller countries.
- 9.2.4 Clusters are descriptive and boundary assignments are uncertain; segment labels should not be treated as immutable identities.
- 9.2.5 Association rules show co-occurrence, not causality. High lift does not prove that recommending one item causes purchase of another.
- 9.2.6 No product margin, stock, promotion exposure, returns cost or customer contact history is available, limiting profit and causal interpretation.
- 9.2.7 Thresholds and business priorities require monitoring as product ranges and customer behaviour change.

9.3 Responsible use

The analysis uses transaction identifiers and behavioural aggregates, not sensitive personal characteristics. CustomerID should remain access-controlled and should not be exposed in campaign reports. Marketing frequency caps, opt-out rights and fairness checks across available non-sensitive operational groups should be enforced. The model should support human decisions rather than automatically deny benefits or impose adverse treatment.

10 Reproducibility and AI Use Transparency

10.1 Reproducible package

Control	Evidence
Dataset identity	UCI source, DOI, metadata and SHA-256 fingerprint
Execution order	Five numbered notebooks from data understanding to association mining
Environment	requirements.txt plus Python and environment records
Determinism	Fixed random seed 42 where applicable; K-Means n_init 20
Data handovers	Validated row counts, schemas, uniqueness assertions and reload checks
Output integrity	Fingerprints for processed datasets, clustering outputs and association outputs
Decision traceability	Thirty-eight logged decisions with evidence and reviewers
Report consistency	Metrics and figures in this report are drawn from versioned documentation outputs

Recommended execution order: 01 data understanding and EDA; 02 data cleaning and preprocessing; 03 customer feature engineering; 04 customer segmentation and clustering; 05 segment-specific association mining. Generated datasets are rebuilt locally from the verified raw file rather than committed as large opaque artifacts.

10.2 AI use declaration

Generative AI tools were used as a support aid for code debugging, explanation, document structuring and language refinement. Group members remained responsible for selecting the business question, validating data assumptions, executing notebooks, reviewing outputs, choosing analytical methods, interpreting results and approving recommendations. AI suggestions were checked against notebook assertions, recorded metrics, output fingerprints and the decision log. No AI-generated claim was accepted solely because it appeared plausible. This use is declared to distinguish assistance from accountable analytical judgement.

11 Conclusion

The project demonstrates a defensible, integrated data-mining workflow. It begins with a precise stakeholder decision, preserves data provenance, creates task-specific datasets, compares a transparent baseline with multiple clustering algorithms, validates stability and converts segment-specific product co-occurrence into practical recommendations. The selected five-cluster K-Means model is not claimed to be mathematically perfect; it is selected because it is stable, balanced enough to operate, interpretable and useful for downstream targeting.

The strongest commercial implication is concentration: Champions account for 53.83% of revenue while representing 8.26% of customers. At the same time, the large Hibernating Low Value group requires cost discipline, and At Risk Regular customers warrant targeted reactivation. The 40 final product rules provide concrete bundle candidates, but value should be confirmed through controlled experiments using incremental revenue, margin and customer-experience measures.

12 References

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Appendix A Notebook and Evidence Map

Notebook	Primary evidence
01 data understanding and EDA	Dataset identity, dictionary, missingness, duplicates, returns, geography and temporal patterns
02 data cleaning and preprocessing	Audited filters, task-specific datasets, handover checks and fingerprints
03 customer feature engineering	RFM and behavioural features, transformations, quality summaries and manifests
04 customer segmentation and clustering	Baseline, 192 experiments, stability, profiles, assignments and figures
05 segment-specific association mining	Sparse baskets, FP-Growth, sensitivity, 2,361 practical rules and 40 recommendations

Appendix B Rubric Coverage Map

Rubric criterion	Report evidence
Business framing and task	Sections 1 and Executive Summary
Workflow and decision log	Section 2 and decision_log.csv
Data understanding and quality	Section 3 and data dictionary/EDA log
Preprocessing and feature engineering	Section 4 and preprocessing/feature summaries
Strategy and comparison	Section 5: baseline, four algorithms and 192 experiments
Evaluation and critical judgement	Section 6: multiple metrics, stability and honest interpretation
Recommendation limitations and value	Sections 7 to 9
Reproducibility and AI transparency	Section 10 and fingerprint outputs